

Global Assembly: Bounded Reduction and Kohler–Jobin Transfer to Sharp Faber–Krahn (Route δ , Plan 2, Building Block GLOBALASSEMBLY)

Abstract

We assemble the final three-stage chain that converts the bounded-class sharp Saint–Venant inequality of the Plan 2 metric route into the unconditional sharp quantitative Faber–Krahn inequality. The chain has three stages: (1) the bounded sharp Saint–Venant inequality $\mathcal{A}(\Omega)^2 \leq C_{B^2}(n, R, \rho_*) \delta_T(\Omega)$ supplied by [6]; (2) the bounded-to-global reduction of [7, Lemmas 5.1–5.3], which internalises the R -dependence at the dimensional radius $R_*(n) := \max\{d(n), 2\}$ and emits a dimensional Saint–Venant constant $c_{SV}^{\text{glob}}(n) > 0$; and (3) the Kohler–Jobin transfer (originally due to Kohler & Jobin 1978; transfer machinery in [8]), which converts sharp Saint–Venant into sharp Faber–Krahn with constant $c_{FK}(n) > 0$. Stages 2 and 3 are cited verbatim from the Plan 1 Final/ library; the distinctive feature of Plan 2 is that the bounded input to stage 2 is delivered by the metric level-set route (no selection principle, no graph entry, no Schauder interpolation, no nearly-spherical theorem).

1 Statement of the main theorem

Throughout $n \geq 2$. We use the standard normalisations: $u_\Omega \in W_0^{1,2}(\Omega)$ with $-\Delta u_\Omega = 1$; $T(\Omega) = \int_\Omega u_\Omega$; $E(\Omega) = -T(\Omega)/2$; $\omega_n = |B_1|$, $L_n = \lambda_1(B_1)$, $T_n = T(B_1) = \omega_n/(n(n+2))$; $\mathcal{A}(\Omega) = \inf_{x_0} |\Omega \Delta(x_0 + B^*)|/|B^*| \in [0, 2)$ (since $|\Omega| = |B^*|$); B^* the ball with $|B^*| = |\Omega|$; $\delta_T(\Omega) := E(\Omega) - E(B^*) \geq 0$.

Theorem 1.1 (Sharp quantitative Faber–Krahn, Plan 2). *There exists a dimensional constant $c_{FK}(n) > 0$ such that for every open $\Omega \subset \mathbb{R}^n$ with $0 < |\Omega| < \infty$,*

$$\left(\frac{|\Omega|}{|B_1|}\right)^{2/n} (\lambda_1(\Omega) - \lambda_1(B^*)) \geq c_{FK}(n) \mathcal{A}(\Omega)^2. \quad (1)$$

The exponent 2 is sharp.

Remark 1.2 (Residual conditionality of the chain). The unconditional Stage 1 statement of [6] depends on the standing gradient hypothesis Hyp-G of [1, §7]: there exist $0 < m_0 \leq M_0 < \infty$ depending only on n, R, ρ_* such that $m_0 \leq |\nabla u_\Omega| \leq M_0$ on $\partial^* E_\rho$ for a.e. $\rho \in [\rho_*, \rho_\delta]$. The upper bound is unconditional via [12]; the lower bound is the residual conditionality of the chain. The bounded reduction class produced by [7] provides the regularity setting in which Hyp-G is expected to hold by a Caffarelli-type non-degeneracy argument; verifying this is the remaining quantitative step in converting the present chain into an unconditional proof of sharp Faber–Krahn. The constants of Theorems 2.1–4.1 depend on $(M_0/m_0)^2$ polynomially.

2 Stage 1: bounded sharp Saint–Venant

Theorem 2.1 ([6]). *For every $n \geq 2$, $R \geq 2$, $\rho_* \in (0, 1)$ there exists $C_{B^2}(n, R, \rho_*) > 0$ such that for every open $\Omega \subset B_R$ with $|\Omega| = \omega_n$,*

$$\mathcal{A}(\Omega)^2 \leq C_{B^2}(n, R, \rho_*) \delta_T(\Omega), \quad (2)$$

equivalently $E(\Omega) - E(B^*) \geq c_{\text{SV}}(n, R, \rho_*) \mathcal{A}(\Omega)^2$ with $c_{\text{SV}} = 1/C_{B^2}$.

This is the Plan 2 metric chain output: `LEVELSETIDENTITY` $\rightarrow$ `METRICFRAMEWORK` $\rightarrow$ `WEIGHTEDMETRICTRACE` $\rightarrow$ `BOUNDARYLAYERTRANSFER`. The centroid and space–time blocks remain auxiliary diagnostics for the alternate centroid route.

Remark 2.2 (Internal parameter ρ_*). We fix $\rho_* = 1/2$ once and for all. Then $c_{\text{SV}}(n, R, 1/2)$ depends only on (n, R) , and we write $c_{\text{SV}}(n, R)$ henceforth. At the dimensional radius $R = R_*(n)$ the constant becomes $c_{\text{SV}}(n, R_*(n))$, which depends only on n ; in contexts where the radius choice is fixed we abbreviate this to $c_{\text{SV}}(n)$.

Remark 2.3 (Distinction from Plan 1). The same bounded inequality is delivered by Plan 1’s `SAINTVENANTASSEMBLY` [9] via selection principle + graph entry + Schauder + nearly-spherical closure. Plan 2 delivers it via the metric level-set chain without those tools.

3 Stage 2: bounded reduction

By [7, Lemma 5.3], every open Ω with $|\Omega| = \omega_n$ can be replaced (without increasing the Faber–Krahn deficit) by an open $\Omega' \subset B_{R_*(n)}$ with $|\Omega'| = \omega_n$; Stage 1 then applies to Ω' .

Theorem 3.1 ([7, Thm. 5.1]). *Suppose (2) holds with constant $c_{\text{SV}}(n, R_*(n))$ at the dimensional radius $R_*(n) := \max\{d(n), 2\}$ where $d(n)$ is the BDV Lemma 5.3 diameter constant. Then for every open $\Omega \subset \mathbb{R}^n$ with $|\Omega| = \omega_n$,*

$$E(\Omega) - E(B_1) \geq c_{\text{SV}}^{\text{glob}}(n) \mathcal{A}(\Omega)^2, \quad (3)$$

with (near-deficit / far-deficit dichotomy of [7, Lemma 5.2])

$$c_{\text{SV}}^{\text{glob}}(n) = \min\{c_{\text{near}}^{\text{glob}}(n), \omega_n^{(n+2)/n} \delta(n)/16\}, \quad (4)$$

and

$$c_{\text{near}}^{\text{glob}}(n) = \frac{\omega_n^{(n+2)/n}}{4C_9(n)(1/\tilde{c}_{\text{SV}}(n) + C_9(n))}, \quad \tilde{c}_{\text{SV}}(n) = \omega_n^{-(n+2)/n} c_{\text{SV}}(n, R_*(n)).$$

By scale invariance of $\mathcal{A}$ (dimensionless) and the homogeneity $\lambda_1(t\Omega) = t^{-2}\lambda_1(\Omega)$, the factor $(|\Omega|/|B_1|)^{2/n}$ on the left and $\mathcal{A}^2$ on the right are both scale-invariant. Replacing Ω by $r\Omega$ with $r := (\omega_n/|\Omega|)^{1/n}$ reduces to $|r\Omega| = \omega_n$, where (3) applies. This gives (5) for arbitrary $|\Omega| < \infty$:

$$E(\Omega) - E(B^*) \geq c_{\text{SV}}^{\text{glob}}(n) \left(\frac{|B^*|}{|B_1|} \right)^{(n+2)/n} \mathcal{A}(\Omega)^2. \quad (5)$$

4 Stage 3: Kohler–Jobin transfer

Theorem 4.1 ([8, Thm. 6.1]). *Suppose (5) holds. Then*

$$\left(\frac{|\Omega|}{|B_1|} \right)^{2/n} (\lambda_1(\Omega) - \lambda_1(B^*)) \geq c_{\text{FK}}(n) \mathcal{A}(\Omega)^2, \quad (6)$$

with

$$c_{\text{FK}}(n) := \min \left\{ \frac{4L_n c_{\text{SV}}^{\text{glob}}(n)}{(n+2)T_n}, \frac{L_n(2^{2/(n+2)} - 1)}{4} \right\} \quad (7)$$

(second branch from Kohler–Jobin large-deficit regime, [8]).

5 Main theorem assembled

Proof of Theorem 1.1. Fix $\rho_* = 1/2$. Apply Theorem 2.1 at $R = R_*(n)$, then Theorem 3.1. By (5) (which follows from Stage 1 + bounded reduction via the scaling step above), apply Theorem 4.1 with $c_{\text{SV}}(n) = c_{\text{SV}}^{\text{glob}}(n)$. The output is (1) with $c_{\text{FK}}(n) > 0$ given by (7).

Sharpness: smooth volume-preserving ellipsoidal perturbations of B^* saturate the exponent 2 on $\mathcal{A}$: see [10, Introduction] and [11, §1] for the explicit family Ω_ε with $\delta(\Omega_\varepsilon) \sim \varepsilon^2$ and $\mathcal{A}(\Omega_\varepsilon) \sim \varepsilon$. $\square$

6 Constants chain

Stage	Constant	Dependence
1 output	$C_{B^2}(n, R, \rho_*)$	n, R, ρ_*
ρ_* fix	$\rho_* = 1/2$	n, R
1 at R_*	$c_{\text{SV}}(n, R_*(n))$	n
2 BDV constants	$C_9(n), \delta(n), d(n)$	n
2 dim. radius	$R_*(n) = \max\{d(n), 2\}$	n
2 output	$c_{\text{SV}}^{\text{glob}}(n)$	n
3 KJ multiplier	$2L_n/((n+2)T_n)$	n
3 output	$c_{\text{FK}}(n)$	n

Verdict: All R, ρ_* -dependence absorbed at the Stage 1→2 interface. Final $c_{\text{FK}}(n)$ depends on n alone.

7 Normalisation conventions

Convention	Choice
Torsion energy sign	$E(\Omega) = -T(\Omega)/2$
Saint–Venant deficit	$\delta_T(\Omega) = E(\Omega) - E(B^*)$
Torsion deficit	$\delta_{\text{SV}} = T(B^*) - T(\Omega) = 2\delta_T$
Volume normalisation	$ \Omega = B^* $, proofs at $ \Omega = \omega_n$
Fraenkel asymmetry	$\mathcal{A}(\Omega) = \inf_{x_0} \Omega \Delta (x_0 + B^*) / B^* $
Scale-invariant deficit	$D(\Omega) = E \Omega ^{-(n+2)/n} - E(B_1)\omega_n^{-(n+2)/n}$
Volume radius	$ E_\rho = \omega_n \rho^n$
Internal ρ_*	$\rho_* = 1/2$
Working radius	$R_*(n) = \max\{d(n), 2\}$
Spectral output	$(\Omega / B_1)^{2/n}(\lambda_1(\Omega) - \lambda_1(B^*))$

The factor 4 in $4L_n c_{\text{SV}}^{\text{glob}}/((n+2)T_n)$ absorbs the factor 2 between δ_{SV} and δ_T .

8 Compatibility with Plan 1

Shared. Stages 2 and 3 are Plan 1 Final/ blocks [7, 8] used verbatim.

Different. Plan 2 delivers Stage 1 via the repaired metric level-set chain [1, 2, 5, 6], with [3, 4] retained as auxiliary centroid-route blocks. The Stage 1 chain uses:

- No selection principle.

- No graph entry.
- No Schauder interpolation.
- No nearly-spherical closure theorem.

This is the methodological contribution of Plan 2.

9 Status: closed-loop summary

Conditional on the repaired Stage 1 building blocks delivering their stated output, the present block closes unconditionally. The smallness regime restriction is absorbed into Stage 2's near-deficit / far-deficit dichotomy.

The seven building blocks of Plan 2:

1. LEVELSETIDENTITY;
2. METRICFRAMEWORK;
3. WEIGHTEDMETRICTRACE;
4. BOUNDARYLAYERTRANSFER;
5. CENTROIDBOUND (auxiliary centroid route);
6. SPACETIMEIDENTITY (auxiliary centroid route);
7. GLOBALASSEMBLY (this block).

References

- [1] Plan 2 building block LEVELSETIDENTITY.
- [2] Plan 2 building block METRICFRAMEWORK.
- [3] Plan 2 building block CENTROIDBOUND.
- [4] Plan 2 building block SPACETIMEIDENTITY.
- [5] Plan 2 building block WEIGHTEDMETRICTRACE.
- [6] Plan 2 building block BOUNDARYLAYERTRANSFER.
- [7] Plan 1 Final/ block BOUNDEDREDUCTION. *Final/BoundedReduction.tex*.
- [8] Plan 1 Final/ block KOHLERJOBINTRANSFER. *Final/KohlerJobinTransfer.tex*.
- [9] Plan 1 Final/ block SAINTVENANTASSEMBLY.
- [10] L. Brasco, G. De Philippis, in *Shape Optimization and Spectral Theory*, De Gruyter Open, 2017.
- [11] M. Allen, D. Kriventsov, R. Neumayer, *Sharp quantitative Faber–Krahn inequalities and the Alt–Caffarelli–Friedman monotonicity formula*, *Ars Inveniendi Analytica* (2023).
- [12] A. Cianchi, V. G. Maz'ya, *Global Lipschitz regularity for a class of quasilinear elliptic equations*, *J. Eur. Math. Soc.* **13** (2011), 247–292.